

Element A: Presentation and Justification of the Problem

Dogs are getting into indoor trash cans, making messes, creating health risks, and causing inconvenience for dog owners and people inside the home. This problem involves dog owners, the dogs themselves, and other people inside the home such as Mr. and Mrs. Larson. Pet companies that make products for dogs can also be our stakeholder. This happens inside the house nearly every day. Dogs dig in the trash because they are attracted to the food and smells inside of the trash cans. This is a small but persistent issue that leaves owners irritated.

Trash cans in homes are used to store food waste and other garbage, which produce strong smells that attract dogs. When dogs get into the trash, they pull the items out and spread the garbage all over the home, creating a mess and possible health hazards for the dogs. The goal of this engineering problem is to create a solution that blocks the dog's access to the trash can while also letting humans operate the trash can with only one hand.

Solving this problem will benefit dog owners and household members by reducing mess, saving time, and preventing dogs from chewing on or eating dangerous items. However, solving this solution can lead to unintended consequences, such as making the trash can harder for people to operate or causing dogs to redirect their behavior somewhere else. This problem relates to other contexts such as using childproof locks and containers to ensure the safety of children.

The consumption of spoiled foods or raw meats have a great chance to cause dogs illness, consuming styrofoam or plastic can lead to gastrointestinal obstruction (Clinic, 2019). Trash cans can contain harmful bacteria such as salmonella, E. coli, and

listeria that can be transferred to humans and animals as well (*The Dirty Truth about Your Trash Bins*, 2025). Eating garbage can cause dogs to get garbage gut which can lead to vomiting, diarrhea, and other symptoms, as well as potentially life threatening complications (*Garbage Gut in Dogs (Garbage Toxicosis)*, n.d.). One of our stakeholders, Ms. L, was ecstatic to hear that we are trying to solve this problem. She was happy to hear that we are putting the owner and dogs safety first. She thinks this would be a great problem to solve so dogs are healthier and less likely to get sick.

Solving this problem is important because dogs getting into indoor trash can cause health risks. Eating spoiled foods can make dogs sick, and swallowing materials like plastic can lead to blockages. Trash cans may also contain harmful bacteria from raw meats such as salmonella and E. Coli which can spread illness if dogs take the meat out of the trash. Because this can happen often and become serious quickly, it is an important problem to solve.

References:

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